WORKED SOLUTION

STEP I 1991, Question 4

Pure • Optimisation in Geometry

STEP I 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q4

The key idea

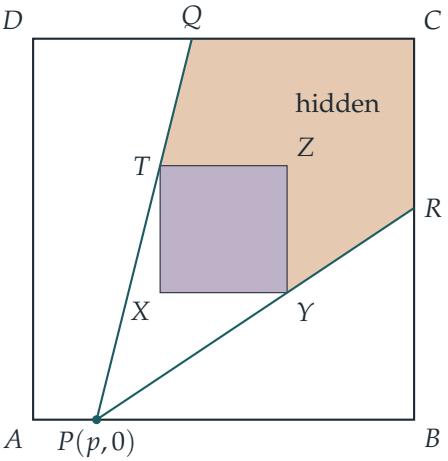
The guard sees every point of the courtyard except those hidden behind the tower, so following the hint we work out the area of the tower's "shadow" as seen from his position $P = (p, 0)$. The shadow is bounded by the two rays from P that just graze the tower. Find where those rays meet the walls, compute the shadow's area as a polygon, and the problem becomes maximising a function of one variable on $[0, 4]$. Remember to check the endpoints.

Solution

Setting up

The courtyard is the big square minus the tower, with area $144 - 16 = 128$. Let the guard be at $P = (p, 0)$ with $0 \leq p \leq 4$, and let $H(p)$ be the area he *cannot* see. The visible area is $128 - H(p)$, so the visible area is smallest when H is largest, and the other way round.

Because $p \leq 4$, P lies below the line $y = 4$ and to the left of (or on) the line $x = 4$. So from P the tower presents its bottom face XY and its left face XT, and its outline is fixed by the corners $T(4, 8)$ and $Y(8, 4)$. The corner $Z(8, 8)$ lies inside the angle TPY and is hidden. The hidden region lies beyond the path $T \to Z \to Y$, between the rays PT and PY.



Where the grazing rays meet the walls

The ray from P through $T(4,8)$ rises 8 for every $4-p$ across, so it reaches $y=12$ at

$$x = p + \tfrac{12}{8}(4-p) = 6 - \tfrac{p}{2}, \qquad \text{giving } Q = \left(6-\tfrac{p}{2},\ 12\right).$$

Since $4 \le 6 - \frac{p}{2} \le 6$, this point really is on the top wall DC.

The ray from P through $Y(8,4)$ rises 4 for every $8-p$ across, so it reaches $x=12$ at

$$y = \frac{4(12-p)}{8-p}, \qquad \text{giving } R = \left(12,\ \frac{4(12-p)}{8-p}\right).$$

This value increases from 6 to 8 as p goes from 0 to 4, so R is on the right wall BC.

The hidden area

The hidden region is the polygon $TQCRYZ$. It is easiest to find its area as

$$H = [PRCQ] - [PYZT],$$

where $[\cdots]$ denotes area: the whole wedge between the two rays, out to the walls, minus the part of the wedge in front of the far faces of the tower (the visible ground in front of the tower together with the tower itself).

Using the shoelace formula for $P(p,0)$, $R(12,q)$, $C(12,12)$, $Q(r,12)$, with $q = \frac{4(12-p)}{8-p}$ and $r = 6 - \frac{p}{2}$:

$$\begin{aligned}[PRCQ] &= \tfrac{1}{2}(pq + (144-12q) + (144-12r) - 12p) \\ &= \tfrac{1}{2}(288 - 12p - 12r - (12-p)q) = 108 - 3p - \frac{2(12-p)^2}{8-p},\end{aligned}$$

using $12r = 72 - 6p$. Similarly, for $P(p,0)$, $Y(8,4)$, $Z(8,8)$, $T(4,8)$:

$$[PYZT] = \tfrac{1}{2}(4p + 32 + 32 - 8p) = 32 - 2p.$$

Therefore

$$H(p) = 76 - p - \frac{2(12-p)^2}{8-p}.$$

Ask yourself Is there a substitution that makes this easy to optimise?

The awkward denominator suggests putting $u = 8 - p$, so that u runs over $[4, 8]$ and $12 - p = u + 4$. Then

$$H = 76 - (8-u) - \frac{2(u+4)^2}{u} = 68 + u - \frac{2u^2 + 16u + 32}{u} = 52 - \left(u + \frac{32}{u}\right).$$

Part (i): smallest visible area

We need H as large as possible, which means $u + \frac{32}{u}$ as small as possible. By the AM–GM inequality, for $u > 0$,

$$u + \frac{32}{u} \;\geq\; 2\sqrt{32} = 8\sqrt{2},$$

with equality exactly when $u = \frac{32}{u}$, that is $u = 4\sqrt{2}$. This lies in $[4, 8]$ (as $4\sqrt{2} \approx 5.66$), so it is allowed. Then $p = 8 - u$, and the visible area is smallest when the guard is

$$\boxed{p = 8 - 4\sqrt{2} \;\approx 2.34 \text{ units from } A}$$

At this point he cannot see $52 - 8\sqrt{2}$, so he sees $76 + 8\sqrt{2} \approx 87.3$ square units.

Part (ii): largest visible area

Now we need H as small as possible, so $f(u) = u + \frac{32}{u}$ as large as possible on $[4, 8]$. Since $f'(u) = 1 - \frac{32}{u^2}$, f is decreasing on $[4, 4\sqrt{2}]$ and increasing on $[4\sqrt{2}, 8]$, so its maximum is at an endpoint. We have

$$f(4) = 4 + 8 = 12, \qquad f(8) = 8 + 4 = 12,$$

which are equal. So H takes its least value, $52 - 12 = 40$, at both $u = 8$ and $u = 4$, that is at $p = 0$ and $p = 4$. The visible area is largest, $128 - 40 = 88$ square units, when the guard stands

$$\boxed{\text{at } A \ (p = 0) \text{ or at } E \ (p = 4)}$$

and at no other point of AE.

Checks

- At $p = 0$ we can do the shadow by hand: the rays hit the walls at $(6, 12)$ and $(12, 6)$, and the hidden hexagon has area 40, matching $H(0)$.

- At $p = 4$ the ray through T runs straight up the line $x = 4$, and $H(4) = 40$ again. The equal endpoint values are a good sign: the formula is not lopsided.
- We counted hidden points on a 600×600 grid, testing whether the line of sight from P passes through the tower. At $p = 8 - 4\sqrt{2}$ this gives 40.687, against $52 - 8\sqrt{2} = 40.686$. At $p = 1$ it gives 40.429, against $H(1) = \frac{283}{7} \approx 40.429$.

Common slips

- Using Z instead of T or Y as a grazing point. From below-left, the tower's outline is fixed by the two corners T and Y, not by the far corner.
- Stopping at the stationary point for part (ii). The stationary point gives the *minimum* visible area, so the maximum must come from the endpoints, and here there are two.

Beyond STEP

This is a small example of a *visibility* problem, the kind studied in computational geometry (the "art gallery problem" asks how many guards are needed to watch every point of a polygonal room). The optimisation step shows a basic fact from analysis: a continuous function on a closed interval attains its maximum and minimum, either at a stationary point or at an endpoint. That is the extreme value theorem, proved carefully in Abbott, *Understanding Analysis*.

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